VIETNAM NATIONAL UNIVERSITY, HO CHI MINH CITY
UNIVERSITY OF TECHNOLOGY
FACULTY OF COMPUTER SCIENCE AND ENGINEERING



DEEP LEARNING IN COMPUTER VISON

Exercise E1–E3

Advisor: Dr. Le Thanh Sach
Dr. Nguyen Quoc Minh

Student: Nguyễn Thị Linh Trang - 2570344
Nguyễn Lê Hoàng Hưng - 2570214

Ho Chi Minh City, November 2026

Contents

1 E1 — Image Classification: Softmax, MLP, CNN **2**
1.1 Training 2
1.2 Architecture 3
1.3 Results (test, seed 0) 3
1.4 Remarks 3
1.5 Error Analysis 4
1.6 Conclusion 6

General Information

Dataset: Fashion-MNIST (28×28 grayscale images, 10 classes), shared across E1–E3.

0.0.0.1 Choosing Fashion-MNIST over MNIST and CIFAR-10:

- **Compared to MNIST:** The MNIST dataset (consisting of handwritten digits from 0–9) is "over-saturated." Models like MLP and CNN can easily achieve around 98–99% accuracy on this dataset. Therefore, if MNIST were used, comparisons regarding architecture or representation methods would be compressed into a very small margin (only a few percent), making the differences between models difficult to observe.
- **Strengths of Fashion-MNIST:** The dataset provides an appropriate level of complexity that helps clearly distinguish the performance between models.
  - In exercise E1, Fashion-MNIST demonstrates a clear gap in accuracy among Softmax, MLP, and CNN.
  - In exercise E2, it allows for comparing the effectiveness of different tokenization methods.
  - In exercise E3, it creates conditions to compare the performance between RNN (LSTM/GRU) and CNN.

0.0.0.2 Data Splitting:

60,000 training images → 55,000 train / 5,000 validation (fixed by seed), 10,000 test images used only once at the end.

0.0.0.3 Preprocessing:

Divide by 255, normalize with mean 0.2860 / std 0.3530 (based on training set statistics). No data augmentation is used to purely compare architectures.

0.0.0.4 Reproducibility:

Seed 0 (determines the split, initialization, and batch order). Executed on CPU or GPU via Colab.

---

1 E1 — Image Classification: Softmax, MLP, CNN

1.1 Training

Data and split: See the general section at the beginning of the report. Custom training loop written from scratch (`forward` → `loss` → `backward()` → `optimizer.step()`), using cross-entropy loss (log-softmax + NLL).

Configuration for each model:
Full data logs are available in ***`outputs/train.log`****.*

Table 1: Training configuration for Softmax, MLP, and CNN.

Model	Optimizer	LR	Batch	Epoch	Regularization
Softmax	Adam	1e-2	128	10	–
MLP	Adam	1e-3	128	10	Dropout 0.2
CNN	Adam	1e-3	128	10	BatchNorm, Dropout 0.2

1.2 Architecture

Table 2: Architecture and parameter count for Softmax, MLP, and CNN.

Model	Architecture	Parameters
Softmax	Flatten(784) → Linear(784→10) → softmax	7,850
MLP	Flatten(784) → Linear(256) → ReLU → Dropout 0.2 → Linear(10)	203,530
CNN	2 × [Conv3×3 - BatchNorm - ReLU - MaxPool2] (32, 64 channels, 28→14→7) → Flatten(3136) → Dropout 0.2 → Linear(10)	50,282

1.3 Results (test, seed 0)

Table 3: Test results for Softmax, MLP, and CNN (seed 0).

Model	Parameters	Best Val Acc	Test Acc	Test Loss	Train Time
Softmax	7,850	83.36%	82.03%	0.520	6 s
MLP	203,530	89.04%	88.86%	0.316	10 s
CNN	50,282	92.12%	91.00%	0.255	192 s

1.4 Remarks

- **Capacity vs. Accuracy:** Softmax is a linear model, so it is capped at around 82%. The model cannot represent non-linear boundaries between classes, by adding a non-linear hidden layer (MLP), the accuracy increases by 6.64% (from 82.03% to 88.67%).

  Meanwhile, CNN has only 50,282 parameters, about 1/4 of the MLP, yet its accuracy is 2.33% higher (91.00% compared to 88.67%). Thanks to the weight sharing of filters and pooling, CNN exploits the 2D structure (local connectivity and translational invariance) that the flatten operation destroys. Therefore, accuracy does not increase proportionally with the number of parameters, but rather depends on whether the architecture fits the data format. However, CNN is slower than MLP on the CPU despite having fewer parameters, because convolutions are more time-consuming.

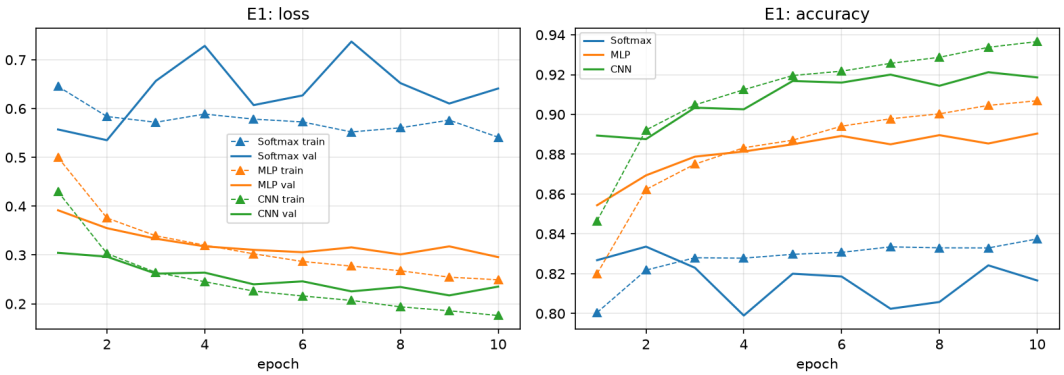


Figure 1: curves.png: Training and validation accuracy/loss curves for Softmax, MLP, and CNN (seed 0).

- **Fit / Overfit:** Softmax almost does not overfit (train ≈ val), but it underfits: loss remains high, and bias is large. MLP begins to separate train and val around epochs 5–6.

  CNN shows the clearest overfitting at the end: train acc (93.66%) vs. val acc (91.86%). Val loss bottoms out around epoch 9 and then increases at epoch 10. Future improvement directions: data augmentation (horizontal flip, slight translation) or early stopping. In this experiment, 10 epochs are kept constant for all three models for a fair comparison.

1.5 Error Analysis

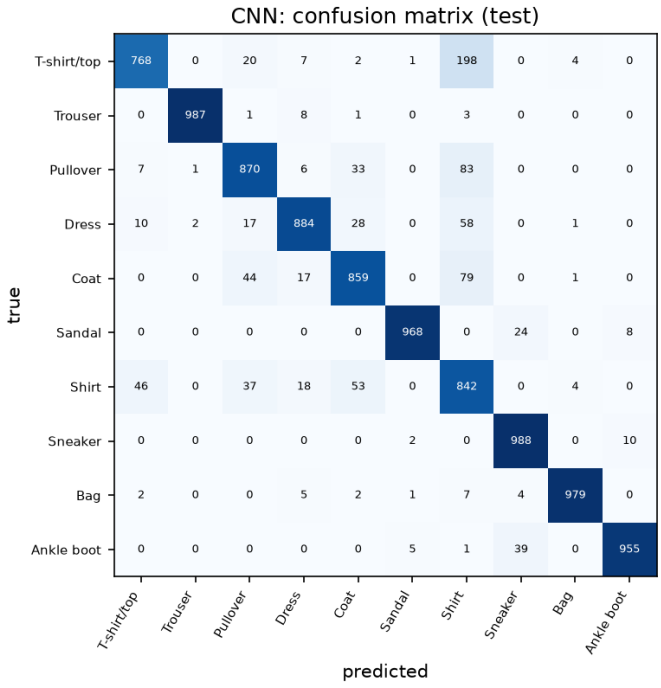


Figure 2: Confusion matrix for CNN (seed 0).

Accuracy by class:



Figure 3: Errors for CNN (seed 0).

Table 4: Accuracy by class for Softmax, MLP, and CNN (seed 0).

Class	**Softmax (%)**	**MLP (%)**	**CNN (%)**
T-shirt/top	86.0	85.1	76.8
Trouser	96.1	96.7	98.7
Pullover	81.7	83.6	87.0
Dress	71.0	92.0	88.4
Coat	71.9	80.1	85.9
Sandal	97.3	94.3	96.8
Shirt	46.3	66.1	84.2
Sneaker	81.6	95.6	98.8
Bag	93.6	97.1	97.9
Ankle boot	94.8	96.1	95.5

Most confused pairs (test set):

Table 5: Most confused pairs (test set).

Model	**Most Confused Pairs**
Softmax	Coat → Pullover: 194, Shirt → T-shirt/top: 190, Shirt → Pullover: 188
MLP	Shirt → T-shirt/top: 128, Coat → Pullover: 105, T-shirt/top → Shirt: 92
CNN	T-shirt/top → Shirt: 198, Pullover → Shirt: 83, Coat → Shirt: 79

- Errors are almost entirely concentrated in the upper-body clothing group (T-shirt, Shirt, Pullover, Coat). These four classes share the same overall shape, differing only in minor details (collar, buttons, sleeve length, zipper). At 28×28 resolution, these details occupy very few pixels.
- Shirt is the most difficult class and also the one where CNN shows the strongest improvement: 46.3% → 66.1% → 84.2% (+37.9 percentage points compared to Softmax). Collars and buttons are local features, which convolutional filters capture much better than a global linear projection.
- CNN shifts the errors rather than just reducing them: it pulls Shirt accuracy up but drops in T-shirt (86.0% → 76.8%) because it leans toward predicting Shirt for plain short-sleeved shirts. T-shirt → Shirt is the most confused pair for CNN (198 images). Overall, it is still a net gain because the increase in Shirt outweighs the decrease in T-shirt.

- Classes with distinct shapes (Trouser, Bag, Sandal, Ankle boot) already achieved high accuracy even with the linear model (around 93.6–97.3%). Sneaker, however, was low in Softmax (81.6%) and strongly improved by CNN (98.8%). Therefore, the majority of the improvement from MLP/CNN comes from the shirt/top group.
- Reviewing `errors_CNN.png`: the majority of the remaining error samples are images where the label itself is ambiguous (a plain long-sleeved shirt could be a Shirt / Pullover / Coat), so part of the error rate is due to the limits of the labeling and resolution, not just the model.

1.6 Conclusion

Test accuracy: Softmax (82.0%) $<$ MLP (88.7%) $<$ CNN (91.0%). The first performance jump comes from non-linearity, the second jump comes from the spatial inductive bias of convolutions, achieved with roughly 4 times fewer parameters than the MLP. The remaining errors are concentrated in visually similar clothing classes, especially Shirt / T-shirt.